Unmasking Parkinson's: Integrating Genetic, Non-Motor, and Biological Markers to Predict Early Disease Risk

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I. INTRODUCTION

Parkinson's Disease (PD) is traditionally diagnosed based on motor symptoms such as tremors, rigidity, and bradykinesia. However, by the time these clinical signs appear, a significant portion of dopaminergic neurons in the substantia nigra has already been lost. This diagnostic lag highlights a critical need for identifying non-invasive biomarkers that can detect PD in its "Prodromal" phase, a period where neurodegeneration has begun but classic motor symptoms are absent.

Recent literature suggests that non-motor symptoms, particularly olfactory dysfunction (hyposmia) and Rapid Eye Movement Sleep Behavior Disorder (RBD), may precede motor onset by years. Furthermore, genetic variants such as LRRK2 and GBA are known to influence disease risk and phenotype, though their specific impact on early non-motor symptoms varies.

The aim of this study is to utilize the Parkinson's Progression Markers Initiative (PPMI) dataset to: (1) determine if non-motor symptoms can distinguish high-risk Prodromal individuals from healthy controls; (2) analyze the phenotypic heterogeneity among genetic subtypes of PD; and (3) validate these symptoms against objective biological markers of brain dopamine levels.

II. METHODS & MATERIALS

A. Data Set

This study utilized data from the Parkinson's Progression Markers Initiative (PPMI), a large-scale, comprehensive observational study. The dataset (PPMI_Curated_Data_Cut_Public_20251112) was obtained from the LONI Image & Data Archive. Data processing and statistical analysis were performed using R Studio.

B. Study Population

I filtered the dataset to include only Baseline (BL) visits to capture early disease characteristics without the confounding effects of long-term medication. The study population was restricted to three cohorts:

1. Healthy Controls (HC): Individuals with no neurological disorders.
2. Parkinson's Disease (PD): Individuals with a clinical diagnosis.
3. Prodromal: Individuals at high risk for PD (e.g., exhibiting hyposmia or RBD) but without a clinical PD diagnosis.

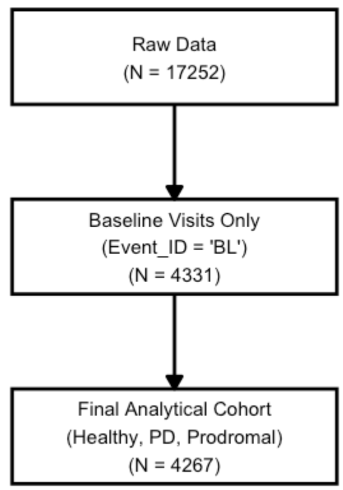

Figure 1: Flow Chart of Data Collection

Table 1: Descriptive Statistics of Study Measurements

Group	Measurement	N	Mean	SD	Median	Min	Max
Healthy Control	Age	336	62.12	11.37	63.77	30.41	85.34
Healthy Control	Autonomic (SCOPA)	334	6.12	4.17	5.00	0.00	29.00
Healthy Control	Cognition (MoCA)	336	27.83	1.70	28.00	20.00	30.00
Healthy Control	Dopamine (DaTscan)	322	1.55	0.31	1.50	0.90	2.60
Healthy Control	Motor (UPDRS III)	332	1.49	2.31	0.00	0.00	13.00
Healthy Control	Sleep (RBD)	336	2.55	2.39	2.00	0.00	13.00
Healthy Control	Smell (UPSIT)	332	33.44	5.51	35.00	10.00	40.00
Prodromal	Age	2456	67.11	6.78	67.06	26.44	91.41
Prodromal	Autonomic (SCOPA)	2446	10.52	6.74	9.00	0.00	43.00
Prodromal	Cognition (MoCA)	2440	26.59	2.51	27.00	11.00	30.00
Prodromal	Dopamine (DaTscan)	0	NaN	NA	NA	Inf	−Inf
Prodromal	Motor (UPDRS III)	2408	4.61	5.76	3.00	0.00	48.00
Prodromal	Sleep (RBD)	2442	5.60	4.09	5.00	0.00	13.00
Prodromal	Smell (UPSIT)	2441	23.59	8.24	23.00	2.00	40.00
PD	Age	1475	62.83	9.84	63.89	28.35	85.25
PD	Autonomic (SCOPA)	1452	10.81	7.06	10.00	0.00	51.00
PD	Cognition (MoCA)	1470	26.72	2.65	27.00	11.00	30.00

C. Statistical Methods

I selected variables representing three domains:

1. Non-Motor Symptoms: Olfactory function was measured using the UPSIT (University of Pennsylvania Smell Identification Test; lower scores indicate worse smell). Sleep disturbances were assessed using the RBDSQ (REM Sleep Behavior Disorder Screening Questionnaire). Autonomic function was measured using SCOPA-AUT.
2. Clinical/Motor: Motor severity was assessed using MDS-UPDRS Part III. Cognitive function was measured using the MoCA (Montreal Cognitive Assessment).
3. Biological/Genetic: Brain dopamine levels were quantified using DaTscan SBR (Specific Binding Ratio) of the Left Putamen (MIA_PUTAMEN_L). Genetic status was categorized into LRRK2, GBA, SNCA, and Sporadic types based on the subgroup variable. The R studio version is 2025.09.0+387

III. RESULTS

A. Non-Motor Symptoms as Early Predictors

Descriptive statistics and visual analysis (Figure 2) revealed distinct profiles for the Prodromal cohort. While Healthy Controls maintained high olfactory scores (Mean UPSIT ~33), the Prodromal group exhibited severe hyposmia (Mean UPSIT ~23.6), statistically indistinguishable from the diagnosed PD group (Mean UPSIT ~22.8). Similarly, sleep disturbance scores (RBDSQ) were highest in the Prodromal group. Importantly, the Prodromal group did not yet exhibit high motor severity scores (UPDRS III), confirming that non-motor symptoms precede motor decline.

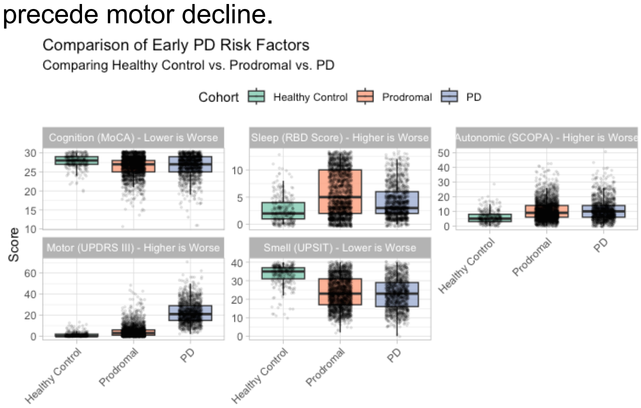

Figure 2: Comparison of Early PD Risk Factors

B. Genetic Heterogeneity in PD Phenotypes

Subgroup analysis of PD patients revealed significant phenotypic differences based on genotype (Figure 3). Patients with GBA mutations exhibited the most severe phenotype, characterized by significant hyposmia and lower cognitive scores. Conversely, patients with LRRK2 mutations preserved olfactory function significantly better than Sporadic and GBA cases, despite having a clinical PD diagnosis (Appendix IV).

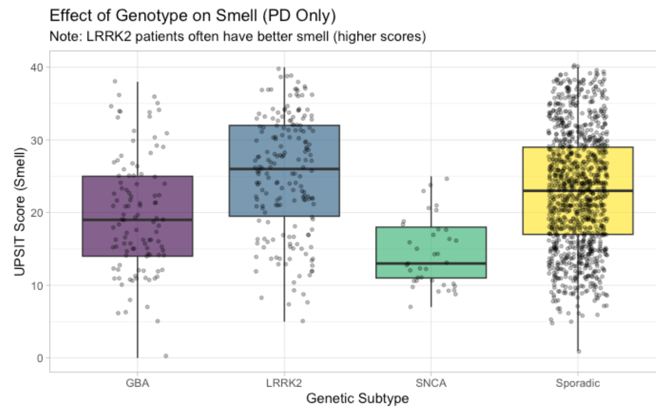

Figure 3: Effect of Genotype on Smell

C. Biological Validation

Correlation analysis demonstrated a significant positive relationship between olfactory scores and striatal binding ratios (Figure 4). A correlation coefficient of $r \approx 0.43$ was observed between UPSIT scores and Left Putamen DaTscan values. This indicates that lower olfactory ability is biologically linked to reduced dopaminergic transporter density in the brain. Conversely, motor symptoms (UPDRS III) showed a negative correlation with dopamine levels, confirming the physiological basis of the motor assessments (Appendix V).

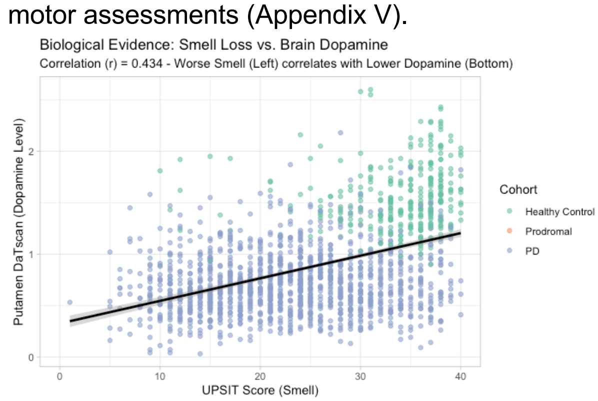

Figure 4: Biological Evidence: Smell Loss vs. Brain Dopamine

IV. CONCLUSIONS

This study confirms that non-motor symptoms, specifically hyposmia and RBD, are potent early indicators of Parkinson's Disease, appearing in the Prodromal phase before significant motor deficits manifest. Furthermore, we demonstrated that "Parkinson's" is not a uniform disease; LRRK2 carriers display a distinct phenotype with preserved smell, while GBA carriers face higher risks of cognitive and olfactory decline. Finally, the strong correlation between smell tests and DaTscan imaging suggests that low-cost olfactory screening could serve as a viable proxy for expensive brain imaging in early screening protocols.

V. DICUSSION

Our findings align with the "dual-hit hypothesis" of PD, where pathology may start in the olfactory bulb and brainstem before reaching the substantia nigra. The preserved smell in LRRK2 carriers suggests a different pathological mechanism, implying that clinical screening criteria should be adjusted based on genetic background. A limitation of this study is its cross-sectional design (baseline only); future work should utilize the longitudinal nature of PPMI data to track the rate of conversion from Prodromal to PD.

ACKNOWLEDGEMENTS

Data used in the preparation of this article were obtained from the Parkinson's Progression Markers Initiative (PPMI) database (www.ppmi-info.org/data). For up-to-date information on the study, visit www.ppmi-info.org.

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Appendix

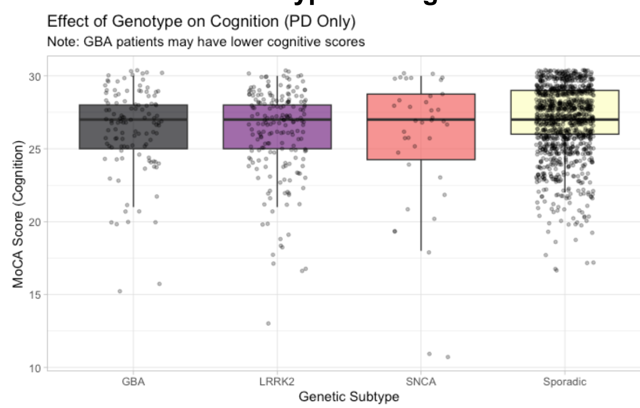
I. Source Data File

The dataset used is PPMI_Curated_Data_Cut_Public_20251112.csv. It contains curated clinical, biological, and imaging data. The specific subset used for analysis includes Baseline visits (EVENT_ID = 'BL') for Cohorts 1, 2, and 4.

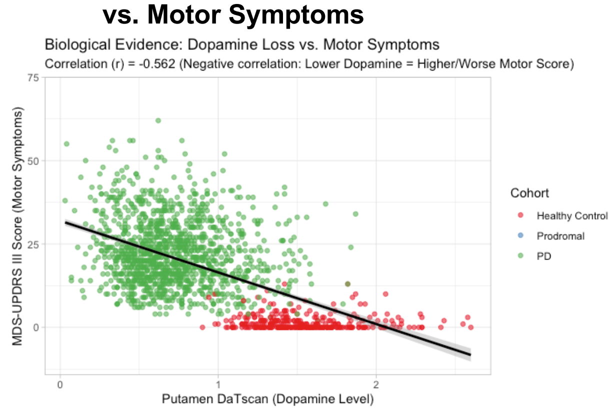
II. Data set

III. R Code

IV. Effect of Genotype on Cognition



V. Biological Evidence: Dopamine Loss vs. Motor Symptoms



VI. PD Risk Factor Odds Ratios